BAISHAKHI SING

Data Science & AI Researcher | Transforming Data into Knowledge, and Knowledge into Innovation

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RESEARCH PROFILE

Bachelor of Technology student in Computer Science and Engineering (Data Science) with research experience in machine learning, explainable AI, and healthcare analytics. My research interests lie at the intersection of Human-Centered AI, Multimodal Learning, and Biomedical Intelligence, with a focus on building AI systems that are reliable, interpretable, and useful in real-world settings. Through research and project experience, I have worked on multimodal intelligence, intelligent systems and healthcare prediction. I aim to pursue an M.S. in Computer Science to deepen my research skills and develop trustworthy AI methods for applications in human well-being, healthcare, and intelligent systems.

EDUCATION

Bachelor of Technology in Computer Science & Engineering **(Data Science)** — 2022 – 2026
The Neotia University |Kolkata, WB, India
GPA: 8.54 / 10

Capstone Project
Nova: Multimodal AI Stress Intelligence Platform

Relevant Coursework:

- Machine Learning
- Deep Learning
- Data Mining & Data Warehousing
- Big Data Analytics (Hadoop & Spark)
- Advanced Data Analytics
- Business Analytics
- Multivariate Analysis, Time Series & Forecasting
- Statistical Analysis using R
- Optimization Techniques
- Data Structures & Algorithms
- Design & Analysis of Algorithms
- Database Management Systems

RESEARCH INTERESTS

- Explainable Artificial Intelligence (XAI)
- Trustworthy Machine Learning
- Multimodal Learning
- Agentic AI
- Computer Vision
- Human-Centered AI
- Graph Neural Network
- Biomedical Intelligence

RESEARCH EXPERIENCE

Research Intern | Indian Institute of Information Technology, Guwahati — Jan 2026-Present

Explainable AI Framework for Early PCOS Prediction Using Lifestyle and Dietary Data

Research Supervisor: Dr. Masihuddin
Research Focus: Explainable AI | Biomedical Machine Learning

Developed an end-to-end explainable machine learning pipeline for early PCOS prediction using lifestyle and dietary data, with a leakage-free workflow covering preprocessing, feature engineering, statistical analysis, and model evaluation to ensure reproducible biomedical AI research. Applied SHAP-based explainability to identify clinically relevant risk factors and improve model interpretability for healthcare decision support.
Research Output: *Manuscript under preparation for submission to a peer-reviewed journal.*

PREPRINTS / MANUSCRIPTS

Manuscripts Under Preparation

- Explainable AI Framework for Early PCOS Prediction Using Lifestyle and Dietary Data
- Nova: A Multimodal AI Framework for Real-Time Stress Intelligence

Research Dataset (Available in Kaggle)
Diabetic Retinopathy Multimodal India
Status: Research dataset.

INDUSTRIAL EXPERIENCE

Project Intern | Defense Research and Development Organization (DRDO), India — Aug 2025 – Dec 2025

- Developed a real-time radar data visualization framework using Python and MATLAB for defence sensor analysis, and optimized radar compilation through algorithmic improvements and system profiling, reducing execution time from 3.66 s to 1.57 s.
- Contributed to **DRDO's internal research documentation**.

Project Intern | DataCore AILABS — Jun 2025 – Jul 2025

- Development an end-to-end MLOps pipeline for retail sales forecasting using Prophet, XGBoost, and LSTM, automating preprocessing, model training, and experiment tracking with MLflow, and deployed a Flask- and Plotly-based dashboard for real-time forecasting and inventory insights.

Project Intern | West Bengal Electronics Industry Development Corporation (Webel) — Jun 2024 – Aug 2024

- Developed and deployed a real-time house price prediction model using Scikit-learn and Pandas, performing data preprocessing, feature engineering, model evaluation, and prototype deployment.

PROJECTS

Nova: Multimodal AI Stress Intelligence Platform ***(Capstone Project)***
Python | PyTorch | Transformers | OpenCV | Speech Processing | Multimodal AI

- Developed an end-to-end multimodal AI platform integrating questionnaire responses, facial expressions, speech, text, and physiological signals for real-time stress assessment, and designed an adaptive multimodal fusion framework with personalized recommendations for mental well-being support.

NeuraTune: Emotion-Based Music Recommender
Python | TensorFlow | OpenCV | CNN | Signal Processing

- Developed an emotion-aware music recommendation system using facial emotion recognition and audio feature extraction. Combined computer vision and affective computing techniques to deliver personalized music recommendations.

Bhasha-Arogya: Cross-Language Healthcare Chatbot
Python | Transformers | NLP | Speech Recognition | Flask | Healthcare AI

- Developed a multilingual healthcare chatbot that translates Bengali patient symptoms into structured clinical information,integrating transformer-based NLP and speech recognition to improve accessibility for AI-assisted healthcare services.

Edge Vision: TinyML for Privacy-Preserving Smart Cities
Python | TinyML | TensorFlow Lite | Edge AI | IoT

- Developed a TinyML-based anomaly detection framework for resource-constrained IoT devices and implemented lightweight edge AI models for privacy-preserving, low-power smart city monitoring.

TECHNICAL SKILLS

- **Programming:** Python, C++, SQL, R, Bash
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, SHAP
- **Deep Learning & Multimodal AI:** CNNs, Transformers, Transfer Learning, Multimodal Learning, Representation Learning
- **Data Science & Analytics:** NumPy, Pandas, Matplotlib, Plotly, Statistical Analysis
- **Computer Vision & NLP:** OpenCV, NLTK, spaCy, Transformer Models, Natural Language Processing
- **MLOps & Deployment:** Docker, MLflow, Git, FastAPI, Flask
- **Edge & Scientific Computing:** Jupyter Notebook, MATLAB, TensorFlow Lite

AWARDS & SCHOLARSHIPS

- Swami Vivekananda Merit-cum-Means Scholarship, Government of West Bengal | 2022–2026
- Tejaswee Merit Scholarship, The Neotia University | 2022–2026
- Winner, National Science Day Poster Presentation (University Level) | 2023
- 1st Runner-up, Shot Put & Discus Throw – The Neotia University Sports Day | Feb 2023

PROFESSIONAL DEVELOPMENT

Certifications

- Elements of AI – University of Helsinki
- Introduction to Generative AI – Google Cloud / Simplilearn
- Machine Learning with Python – freeCodeCamp

Workshops

- Data Visualization Techniques in Business Analytics (Tableau & Power BI) | Nov 2024
- Application of R in Business Analytics (Case Study Approach) | Apr 2024

TEACHING SKILLS

Teaching Assistant – Introduction to Programming (Python)|The Neotia University 2023 – 2024
Assisted middle school students with Basic Python programming and coding assignments.

LANGUAGES

English: Proficient | Bengali: Native | Hindi: Proficient | Korean: Intermediate

REFERENCES

- Dr. Usha Rani Gogoi | ✉ usharani.gogoi@tnu.in
  Head Of the Department, Dept. of Computer Science & Engineering | The Neotia University

- Dr. Ayan Chatterjee | ✉ ayan.chatterjee@tnu.in
  Assistant Professor, Dept. of Basic Science | The Neotia University

**I affirm that the information provided above is accurate and complete to the best of my knowledge*